Series ЄFGHE

Set-5

प्रश्न-पत्र कोड
Q.P. Code **56(B)**

रोल नं.
Roll No.

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें ।
Candidates must write the Q.P. Code on the title page of the answer-book.

रसायन विज्ञान
(केवल दृष्टिबाधित परीक्षार्थियों के लिए)
CHEMISTRY
(FOR VISUALLY IMPAIRED CANDIDATES ONLY)

*निर्धारित समय : **3** घण्टे* | *अधिकतम अंक : **70***

*Time allowed : **3** hours* | *Maximum Marks : **70***

- कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ **23** हैं ।
- प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें ।
- कृपया जाँच कर लें कि इस प्रश्न-पत्र में **35** प्रश्न हैं ।
- **कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में प्रश्न का क्रमांक अवश्य लिखें ।**
- इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है । प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा । 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे ।
- Please check that this question paper contains **23** printed pages.
- Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- Please check that this question paper contains **35** questions.
- **Please write down the serial number of the question in the answer-book before attempting it.**
- 15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.

General Instructions :

Read the following instructions carefully and strictly follow them :

*(i) This question paper contains **35** questions. **All** questions are **compulsory**.*

*(ii) This question paper is divided into **five** Sections – **A, B, C, D** and **E**.*

*(iii) In **Section A** – Questions no. **1** to **18** are multiple choice (MCQ) type questions, carrying **1** mark each.*

*(iv) In **Section B** – Questions no. **19** to **25** very short answer (VSA) type questions, carrying **2** marks each.*

*(v) In **Section C** – Questions no. **26** to **30** are short answer (SA) type questions, carrying **3** marks each.*

*(vi) In **Section D** – Questions no. **31** and **32** are case-based questions carrying **4** marks each.*

*(vii) In **Section E** – Questions no. **33** to **35** are long answer (LA) type questions carrying **5** marks each.*

(viii) There is no overall choice. However, an internal choice has been provided in 2 questions in Section B, 2 questions in Section C, 2 questions in Section D and 2 questions in Section E.

*(ix) Use of calculators is **not** allowed.*

SECTION A

*Questions no. **1** to **18** are Multiple Choice (MCQ) type Questions, carrying **1** mark each.* *$18 \times 1 = 18$*

1. For an electrolyte undergoing dissociation in an aqueous solution the Van't Hoff factor :

(a) is always less than one.

(b) is always greater than one.

(c) has zero value.

(d) has negative value.

2. Which of the following cells has a constant voltage throughout its life ?

(a) Dry cell (b) Electrolytic cell

(c) Mercury cell (d) Daniell cell

3. How many Faradays are required to reduce one mole of Sn^{4+} to Sn^{2+} ?

(a) 2·0 (b) 4·0

(c) 1·0 (d) 6·0

4. For the reaction $A + 2B \longrightarrow 3C + D$, $d[C]/dt$ is equal to :

(a) $-\frac{d[A]}{dt}$ (b) $-\frac{d[B]}{dt}$

(c) $+\frac{3d[A]}{dt}$ (d) $\frac{-3}{2}\frac{d[B]}{dt}$

5. The unit of rate constant and rate of reaction are identical for a :

(a) zero order reaction (b) first order reaction

(c) second order reaction (d) third order reaction

6. Which of the following transition metals do ***not*** show variable oxidation states ?

(a) Cu (b) Ti

(c) Mn (d) Sc

7. The coordination number of Ag in $[Ag(NH_3)_2]Cl$ is :

(a) 3 (b) 1

(c) 2 (d) 4

8. Amongst the following, the most stable complex is :

(a) $[Fe(C_2O_4)_3]^{3-}$ (b) $[FeCl_6]^{3-}$

(c) $[Fe(H_2O)_6]^{3+}$ (d) $[Fe(NH_3)_6]^{3+}$

9. Acetyl chloride is treated with H_2 in the presence of Pd-$BaSO_4$. The product formed is :

(a) CH_3CH_2OH (b) CH_3CHO

(c) CH_3COOH (d) CH_3COCH_3

10. The Gabriel phthalimide synthesis is used for the preparation of :

(a) primary aromatic amines

(b) primary aliphatic amines

(c) secondary amines

(d) tertiary amines

11. The action of nitrous acid on ethylamine gives mainly :

(a) ethyl nitrite (b) ethane

(c) nitroethane (d) ethyl alcohol

12. The helix structure of proteins is stabilized by :

(a) peptide bonds (b) disulphide bonds

(c) hydrogen bonds (d) Van der Waals forces

13. The deficiency of which Vitamin causes rickets ?

(a) Vitamin A (b) Vitamin B

(c) Vitamin C (d) Vitamin D

14. Oligosaccharides on hydrolysis could yield :

(a) 3 to 9 monosaccharides

(b) 4 to 10 monosaccharides

(c) more than 10 monosaccharides

(d) 3 to 10 monosaccharides

For Questions number 15 to 18, two statements are given — one labelled as Assertion (A) and the other labelled as Reason (R). Select the correct answer to these questions from the codes (a), (b), (c) and (d) as given below.

(a) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is ***not*** the correct explanation of the Assertion (A).

(c) Assertion (A) is true, but Reason (R) is false.

(d) Assertion (A) is false, but Reason (R) is true.

15. *Assertion (A) :* Molarity of solution in liquid state changes with temperature.

Reason (R) : The volume of a solution changes with change in temperature.

16. *Assertion (A) :* Fe^{2+} acts as a reducing agent.

Reason (R) : Fe^{3+}state is stable due to $3d^5$ configuration.

17. *Assertion (A) :* Nucleophilic substitution reaction of an optically active halide gives a mixture of enantiomers.

Reason (R) : S_N2 reactions of optically active halides are accompanied by inversion of configuration.

18. *Assertion (A) :* The boiling point of ethanol is lower than that of diethyl ether.

Reason (R) : In ethanol, the molecules are associated due to intermolecular hydrogen bonding, whereas in diethyl ether it is not possible.

SECTION B

19. What type of deviation from Raoult's law is shown by chloroform and acetone solution ? Give reason. *2*

20. (a) Write anode, cathode and overall reaction of lead storage battery. *2*

OR

(b) Define fuel cell with an example. Write two advantages of fuel cell compared to other ordinary cells. *2*

21. Write two differences between order of reaction and molecularity of reaction. *2*

22. (a) Using IUPAC norms, write the systematic names of the following complexes : *2×1=2*

(i) $K_3[Fe(CN)_6]$

(ii) $[Co(en)_2Cl_2]^+$

OR

(b) Write the role of coordination compounds with an example in each of the following : *2×1=2*

(i) Biological system

(ii) Extraction of metals

23. What happens when : *2×1=2*

(a) 1-chlorobutane is treated with alcoholic KOH ?

(b) Chlorobenzene is treated with sodium in dry ether ?

24. Write the chemical equation involved in : *2×1=2*

(a) Reimer-Tiemann reaction

(b) Williamson ether synthesis

25. Give reasons : *2×1=2*

(a) Oxidation of aldehydes is easier than that of ketones.

(b) Carboxylic acid is a stronger acid than phenol.

SECTION C

26. The freezing point of a solution containing 5 g of benzoic acid ($M = 122\ g\ mol^{-1}$) in 35 g of benzene is depressed by 2·94 K. What is the percentage association of benzoic acid if it forms a dimer in solution ? *3*

(K_f for benzene = $4{\cdot}9\ K\ kg\ mol^{-1}$)

27. The rate constant of a reaction is $2 \times 10^{-2}\ s^{-1}$ at 300 K and $8 \times 10^{-2}\ s^{-1}$ at 340 K. Calculate the energy of activation of the reaction. *3*

[$R = 8{\cdot}314\ JK^{-1}\ mol^{-1}$], [$\log 2 = 0{\cdot}3010$, $\log 4 = 0{\cdot}6021$]

28. (a) What is a 'chelate complex' ? Give an example.

(b) Write the hybridisation and magnetic character of $[FeF_6]^{3-}$.
[Atomic number : Fe = 26]

(c) What type of isomerism is shown by the coordination compound $[Cr(H_2O)_5Br]SO_4$? *3×1=3*

29. Answer any ***three*** of the following : *3×1=3*

(a) p-nitrochlorobenzene undergoes nucleophilic substitution reaction faster than chlorobenzene.

(b) Why is (±)-Butan-2-ol optically inactive, though it contains a chiral carbon atom ?

(c) Why is chloroform kept in airtight dark coloured bottle ?

(d) Write the major product formed when 2-Bromobutane is heated with alcoholic KOH.

30. (a) How do you convert the following : *3×1=3*

(i) Phenol to Benzene

(ii) Ethanol to propan-2-ol

(iii) Anisole to 2-methoxyacetophenone

OR

(b) (i) How will you distinguish between primary, secondary and tertiary alcohol by Lucas reagent ?

(ii) Why is o-nitrophenol steam volatile while p-nitrophenol is not ? *2+1=3*

SECTION D

The following questions are case-based questions. Read the case carefully and answer the questions that follow.

31. Amines can be considered as derivatives of ammonia obtained by replacement of hydrogen atoms with alkyl or aryl groups. Like ammonia, all the three types of amines have one unshared electron pair on nitrogen atom due to which they behave as Lewis bases. Amines are usually formed from nitro compounds, halides, amides, etc. They exhibit hydrogen bonding which influences their physical properties. Alkyl amines are found to be stronger bases than ammonia. In aromatic amines, electron releasing and withdrawing groups, respectively increase and decrease their basic character.

Answer the following :

(i) Out of aniline and methylamine, which is a stronger base and why ? *1*

(ii) Why does tertiary amine have lower boiling point than primary amine of comparable molecular masses ? *1*

(iii) What happens when : *2×1=2*

(1) Ethanamide is heated with Br_2 and aqueous KOH ?

(2) Benzene diazonium chloride is treated with Ethanol ?

OR

(iii) Write short notes on the following : *2×1=2*

(1) Carbylamine reaction

(2) Acetylation of aniline

32. Carbohydrates are broadly classified into three groups : monosaccharides, oligosaccharides and polysaccharides. Monosaccharides are held together by glycosidic linkages to form disaccharides or polysaccharides.

Proteins which contain only α-amino acids are called simple proteins. Proteins get denatured if subjected to change in temperature or pH.

Answer the following :

(i) What is the difference between glycosidic linkage and peptide linkage ? *1*

(ii) What is the effect of denaturation on the structures of protein ? *1*

(iii) Define the following terms : *2×1=2*

(1) Essential amino acids

(2) Anomers

OR

(iii) What are the hydrolysis products of : *2×1=2*

(1) Sucrose

(2) Lactose

SECTION E

33. (a) 'A' and 'B' are two functional isomers of compound C_3H_6O. On heating with NaOH and I_2, isomer 'B' forms yellow precipitate of iodoform whereas isomer 'A' does not form any precipitate. Identify 'A' and 'B'.

(b) Give reasons for the following :

(i) Carboxylic acid does not give reactions of carbonyl group.

(ii) Propanone is less reactive towards nucleophilic addition reaction as compared to propanal.

(c) Write a simple chemical test to distinguish between benzoic acid and phenol. *2+2+1=5*

34. (a) (i) Account for the following :

(1) Mn shows the highest oxidation state of +7 with oxygen but with fluorine it shows the highest oxidation state of +4.

(2) Cr^{2+} is a strong reducing agent.

(3) Cu^{2+} salts are coloured while Zn^{2+} salts are white.

(ii) Complete and balance the following chemical equations :

(1) $2MnO_2 + 4KOH + O_2 \xrightarrow{\Delta}$

(2) $Cr_2O_7^{2-} + 14H^+ + 6I^- \longrightarrow$ *3+2=5*

OR

(b) (i) The elements of 3d transition series are given as :

Sc Ti V Cr Mn Fe Co Ni Cu Zn

Answer the following :

(1) Which element has the highest melting point and why ?

(2) Which element is most stable in +2 oxidation state and why ?

(3) Which element has lowest enthalpy of atomisation and why ?

(ii) Write the preparation of $Na_2Cr_2O_7$ from chromite ore. *3+2=5*

35. (a) (i) Calculate E^o_{cell} for the following reaction at 298 K :

$2Al\,(s) + 3Cu^{2+}\,(0{\cdot}01\ M) \longrightarrow 2Al^{3+}\,(0{\cdot}01\ M) + 3Cu\,(s)$

Given : $E_{cell} = 1{\cdot}98$ V, $\log 10 = 1$

(ii) Using the E^o values of A and B, predict which is better for coating the surface of iron ($E^\circ_{Fe^{2+}/Fe} = -\,0{\cdot}44$ V) to prevent corrosion and why ? *3+2=5*

Given : $E^\circ_{(A^{2+}/A)} = -\,2{\cdot}37$ V

$E^\circ_{(B^{2+}/B)} = -\,0{\cdot}14$ V

OR

(b) (i) The conductivity of 0·001 M solution of CH_3COOH is $3{\cdot}905 \times 10^{-5}\ S\ cm^{-1}$. Calculate its degree of dissociation (α).

Given : $\lambda^\circ_{(CH_3COO^-)} = 40{\cdot}9\ S\ cm^2\ mol^{-1}$

$\lambda^\circ_{H^+} = 349{\cdot}6\ S\ cm^2\ mol^{-1}$

(ii) State Faraday's second law of electrolysis.

(iii) What happens if external potential applied becomes greater than E^o_{cell} of electrochemical cell ? *3+1+1=5*